

Equations with Unknowns on Both Sides

AQA · KS3 Year 8

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE

$$2x + 3 = 11$$

$$-3 \text{ both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



An equation balances: do the same to both sides to keep it level.

COMMON MISTAKES

Not doing the move to both sides.

Moving the larger letter term and creating a needless negative.

$5x - 3x = 2x$, not $8x$.

Stopping at $ax = c$ without the final division.

METHOD

- 1 Choose the side to keep the letters
- 2 Subtract the smaller letter term (both sides)
- 3 Collect the numbers using inverses
- 4 Solve the simpler equation
- 5 Check both sides match

WORKED EXAMPLE

Solve $5x + 2 = 3x + 10$.

Subtract $3x$: $2x + 2 = 10$. Subtract 2: $2x = 8$. Divide by 2: $x = 4$.

Check: $5(4) + 2 = 22 = 3(4) + 10$

VISUAL AID

Balance scale with x-tiles and unit-tiles on both pans. Remove 3 x-tiles from each pan, then remove 2 units from each pan, leaving $2x = 8$ $x = 4$.

1. (2 marks)

Solve: a) $5x = 2x + 9$ b) $7x = 3x + 20$

2. (2 marks)

Solve: a) $6x + 1 = 4x + 9$ b) $8x - 5 = 6x + 7$

3. (2 marks)

Solve: a) $9x - 4 = 4x + 11$ b) $4x + 7 = x + 22$

4. (3 marks)

Solve: a) $3x + 8 = 7x$ b) $2x - 5 = 5x + 4$ c) $10 - x = 4 + x$

5. (3 marks)

Solve: a) $5x + 3 = 2x - 9$ b) $7 - 2x = x + 1$ c) $4x - 1 = 6x - 9$

6. (3 marks)

Sara ($4x+2$) sweets = Tom ($6x-8$) sweets. a) Equation. b) Find x . c) How many each?

7. (4 marks)

Plan A: $\pounds 10 + \pounds 2/\text{GB}$. Plan B: $\pounds 4 + \pounds 4/\text{GB}$. a) Equation in g . b) GB where equal. c) That cost.

8. (5 marks)

Rod A ($5x-3$) cm = Rod B ($2x+9$) cm. a) Find x . b) Length of each. c) Rod 5 cm less than twice Rod A.